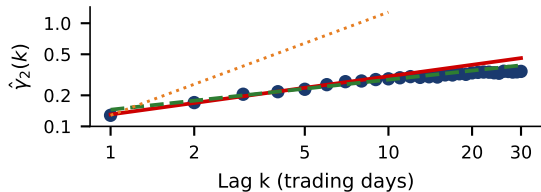
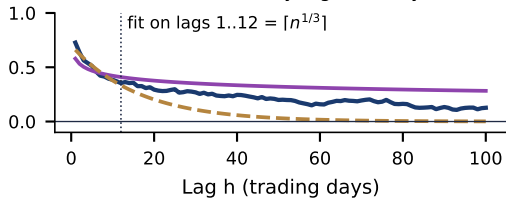


Roughness: variogram of daily log-volatility (log-log)



- Empirical variogram of $\log \hat{\sigma}_t$
- OLS, $k = 1..6$: $\hat{\alpha} = -0.32$
- NLLS fit, $m = 15$; mean over $m = 10..20$: $\hat{\alpha}^* = -0.35$
- ... Brownian roughness, $\alpha = 0$ (slope 1)

Persistence: ACF of daily log-volatility



- Empirical ACF of $\log \hat{\sigma}_t$
- Cauchy class, noise-robust: $\hat{\beta}^* = 0.24$ (long memory)
- Exponential decay (OU, short memory): half-life 11 days